

RELATIONSHIP BETWEEN FORCE-TIME AND VELOCITY-TIME CHARACTERISTICS OF DYNAMIC AND ISOMETRIC MUSCLE ACTIONS

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ABSTRACT

Khamoui, AV, Brown, LE, Nguyen, D, Uribe, BP, Coburn, JW, Noffal, GJ, and Tran, T. Relationship between force-time and velocity-time characteristics of dynamic and isometric muscle actions. *J Strength Cond Res* 25(1): 198–204, 2011—Previous research has investigated the force-time curve characteristics of isometric and dynamic muscle actions; however, few studies have addressed their relationship to dynamic exercise velocity-time variables. The purpose of this study was to investigate relationships between velocity-time characteristics (high pull and vertical jump peak velocity and rate of velocity development [HPPV, HPRVD, VJPV, VJRVD]), force-time characteristics (isometric peak force [IsoPF], body mass adjusted isometric peak force [IsoPF/BM], isometric rate of force development at different millisecond windows [IsoRFD50-250], dynamic peak force [HPPF], body mass adjusted dynamic peak force [HPPF/BM]), and vertical jump height (VJHeight). Nineteen recreationally trained men (age 23.89 ± 2.92 yr; height 176.32 ± 7.06 cm; mass 78.76 ± 16.50 kg) completed 2 testing sessions. The first session consisted of 3 isometric mid-thigh pulls on a force plate with each repetition held for 3 seconds. On the second testing session, subjects completed 3 dynamic mid-thigh high pulls with 30% IsoPF followed by 3 vertical jumps on a force plate. The HPRVD correlated with IsoRFD50 ($r = 0.52$) and IsoRFD100 ($r = 0.49$). The HPPV correlated with IsoPF/BM ($r = -0.60$), IsoRFD50 ($r = 0.56$), and IsoRFD100 ($r = 0.56$). The VJHeight correlated with IsoPF/BM ($r = 0.61$), whereas VJPV correlated with IsoPF/BM ($r = 0.62$). These correlations suggest that explosive isometric force production within 50 to 100 milliseconds may influence the ability to accelerate an implement or body and attain high velocity, albeit in a moderate

fashion. In addition, body mass adjusted strength may positively influence vertical jump parameters.

KEY WORDS high pull, rate of force development, rate of velocity development

INTRODUCTION

Skeletal muscle function can be evaluated by way of force-time records obtained during dynamic and isometric exercise (2,6,10–12,15). Previous investigations have used these force-time records to examine the relationship between measures of muscle function and various parameters of human performance. For instance, peak force and rate of force development have been correlated to sprint cycling (26), jump performance (6,8–10,12,15,24,28), weightlifting outcomes (5,10), and throwing ability (25).

Several studies specifically investigated the force-time characteristics of isometric and dynamic muscle actions (10,12,16). Haff et al. (12) found moderate to strong correlations between mid-thigh pull isometric peak force and high pull dynamic peak force at various loads. They also found similar relationships between the latter and mid-thigh pull isometric rate of force development, leading them to suggest that similar neuromuscular components modulate production of isometric strength, dynamic strength, and explosive isometric strength. McGuigan and Winchester (16) observed strong correlations between mid-thigh pull isometric peak force and dynamic strength assessment in collegiate football players. In similar fashion, Haff et al. (10) evaluated force-time records during dynamic and isometric muscle actions in elite female weightlifters and found strong relationships between the same force measure obtained from a mid-thigh isometric pull and dynamic high pull (i.e., isometric peak force and dynamic peak force, relative isometric peak force, and relative dynamic peak force). Clearly, prior research has addressed the force-time characteristics of dynamic and isometric muscle actions.

In contrast with the availability of studies documenting force-time curve analysis of isometric and dynamic muscle actions, few have examined their relationship to velocity-time

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characteristics of dynamic activities. A handful have investigated the relationship between isometric and dynamic force-time variables and peak velocity attained during the high pull and vertical jump (10,18); however, few, to our knowledge, have investigated other kinematic properties such as acceleration.

The importance of acceleration to human performance can be derived from Newton's second law, which asserts that a mass will accelerate when subjected to a force. Specifically, rate of velocity development, defined as the slope of the velocity-time record and representative of acceleration from 0 to peak velocity, has been a variable of interest in previous studies (4,17). Brown and Whitehurst (4) found enhanced knee extensor rate of velocity development after short-term isokinetic training, whereas Murray et al. (17) observed improved limb acceleration after a similar intervention. This ability to attain peak velocity in a short time period applies to most sports, particularly those requiring individuals to move their mass or an implement quickly.

Because such a parameter contains obvious connections to human performance, it would be useful to acquire a better understanding of its relationship with isometric and dynamic force-time variables. Although this does not establish causation, useful information regarding program design considerations might still be inferred. Therefore, the purpose of this study was to determine the relationship between dynamic exercise velocity-time parameters, force-time characteristics of isometric and dynamic muscle actions, and vertical jump performance.

METHODS

Experimental Approach to the Problem

This study used 2 testing sessions separated by 48 hours recovery to determine the relationship between dynamic exercise velocity-time characteristics, isometric force-time characteristics, dynamic force-time characteristics, and vertical jump performance as measured by peak jump height (VJHeight). Isometric force-time characteristics including peak force (IsoPF), peak force relative to body mass (IsoPF/BM), and rate of force development at various time frames (IsoRFD50, IsoRFD100, IsoRFD150, IsoRFD200, IsoRFD250, IsoRFDMax) were obtained using a mid-thigh isometric pull (10–12,26). Dynamic force-time variables were obtained from a mid-thigh high pull with a 30% IsoPF load (10–12,26) and consisted of high pull peak force (HPPF), high pull peak force relative to body mass (HPPF/BM), and high-pull maximum rate of force development (HPRFDMax).

Velocity-time parameters (peak velocity, rate of velocity development) were derived from 2 different dynamic activities: the mid-thigh high pull with a 30% IsoPF load and the vertical jump. The dynamic high pull provides kinematic aspects of the barbell (external object) generated by the subject, whereas the vertical jump requires movement of one's body mass, thereby encompassing a greater range of dynamic activities. Dynamic exercise velocity-time variables

included high pull peak velocity (HPPV), high pull rate of velocity development (HPRVD), vertical jump peak velocity (VJPV), and vertical jump rate of velocity development (VJRVD).

Subjects

Nineteen recreationally trained men (age 23.89 ± 2.92 yr; height 176.32 ± 7.06 cm; mass 78.76 ± 16.50 kg; IsoPF 1696.50 ± 202.87 N; VJHeight 36.72 ± 7.99 cm) completed 2 testing sessions separated by 48 hours rest. All subjects signed an informed consent document approved by the university institutional review board before participating in the study. Subjects did not have a history of musculoskeletal injuries within the past year and were instructed to refrain from performing lower-body exercises for the duration of the study. A certified strength and conditioning specialist monitored each testing session.

Procedure

The first testing session began with a 5-minute warm-up on a cycle ergometer at a self-selected cadence with a 25 W load. After the warm-up, subjects performed three 3-second long maximum isometric mid-thigh pulls inside a power rack while standing on a force plate. To ensure the device remained static, investigators clamped the barbell to the crash bars of the power rack (10–12). The force plate (Advanced Mechanical Technology, Inc., Watertown, MA, USA) sampled at 1,000 Hz and interfaced with a personal computer containing an A/D converter and Labview software (LabVIEW, version 7.1, National Instruments Corporation, Austin, TX, USA), which provided values for peak force and rate of force development at 50, 100, 200, and 250 milliseconds and maximum during the isometric mid-thigh pull. Rate of force development was calculated from the slope of the force-time curve (i.e., $\Delta\text{Force}/\Delta\text{Time}$) using peak force and the elapsed time between 0 and peak force as values (Figure 1). Rate of force development at 50 to 250 milliseconds was determined by dividing force at the time of interest after contraction initiation by the corresponding time (e.g., force at 50 ms/50 ms). Force signals did not undergo filtering before further analyses, and the absence of data smoothing may be a potential limitation when calculating these derivatives.

The second testing session began with the same 5-minute warm-up followed by 3 dynamic mid-thigh high pulls on a force plate at 30% IsoPF. Assessment of the dynamic high pulls involved the placement of 2 linear velocity transducers adjacent to the bar collars (Model V-80-L7M, Unimeasure, Inc., Corvallis, OR, U.S.A.). The transducers also interfaced with a personal computer containing an A/D converter and LabVIEW software, which averaged values from both devices to provide HPPV. Values for HPRVD were determined by dividing HPPV by the elapsed time between 0 and peak velocity (Figure 2). During isometric and mid-thigh pull assessments, all subjects used similar knee angles for both exercises (127–145°) and were instructed to pull as hard

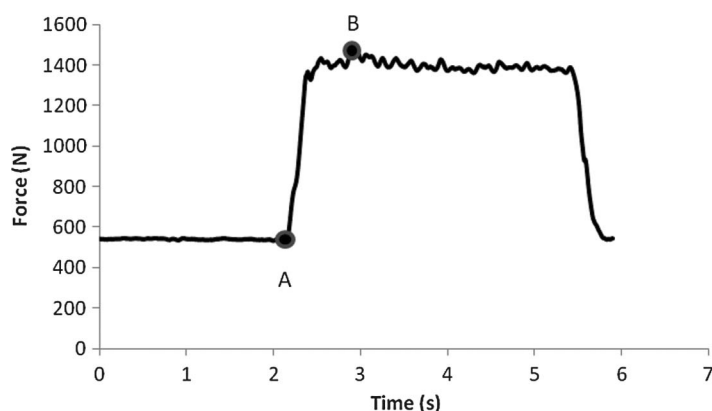


Figure 1. Typical force-time record obtained during isometric mid-thigh pull. Point "A" represents onset of rise in force, whereas point "B" indicates maximum force expression. Rate of force development was calculated as slope of segment between "A" and "B."

and quickly as possible with 2 minutes recovery provided between trials (11).

Upon completion of the dynamic high pulls, subjects performed 3 countermovement vertical jumps with arm swing to a self-determined depth and were consistently encouraged to jump as high as possible. All jumps were performed on a multicomponent AMTI force platform that interfaced with a personal computer at a sampling rate of 1,000 Hz. Data acquisition software collected values for vertical ground reaction force. The VJPV was determined by subtracting body weight from the force-time curve, dividing by body mass, and integrating with respect to time using the trapezoidal rule for numerical integration. The VJHeight was calculated by way

of the following equation of uniformly accelerated motion: $VJHeight = v^2/2g$, where v and g represent vertical velocity at take-off and gravitational acceleration, respectively. Take-off was defined as the instant force dropped below 2.2 kg. The VJRVD resulted from identifying peak velocity, time at peak velocity, and time at 0 velocity to calculate the slope of the velocity-time curve (Figure 3). Specifically, VJPV was divided by the elapsed time between 0 and peak velocity and can be represented by the equation $VJRVD = VJPV / (T_{peak} - T_{zero})$, where T_{peak} and T_{zero} represent the time at peak velocity

Statistical Analyses

Peak values for each dependent variable were used for statistical analysis. Pearson product moment correlation coefficients described the relationship between dependent variables with an a priori alpha level of 0.05. We excluded assessment of the relationship between velocity-time and force-time characteristics of the high pull (HPPV, HPRVD, HPPF, HPPF/BM, HPRFDMax) on the basis of those measures being derived from the same exercise. Post hoc power analysis revealed a mean statistical power of 0.80 and a range of 0.64 to 0.90 for significant correlations. These values closely resemble those obtained in a similar study by Nuzzo et al. (18), who reported average power as 0.83 with a range of 0.56 to 1.00 for significant correlations. Intraclass correlation coefficients demonstrated acceptable day-to-day reliability for all dependent variables ($R = 0.75-0.99$). With the exception of maximum RFD values (IsoRFDMax [$R = 0.75$] and HPRFDMax [$R = 0.84$]), all dependent variables obtained $R \geq 0.94$.

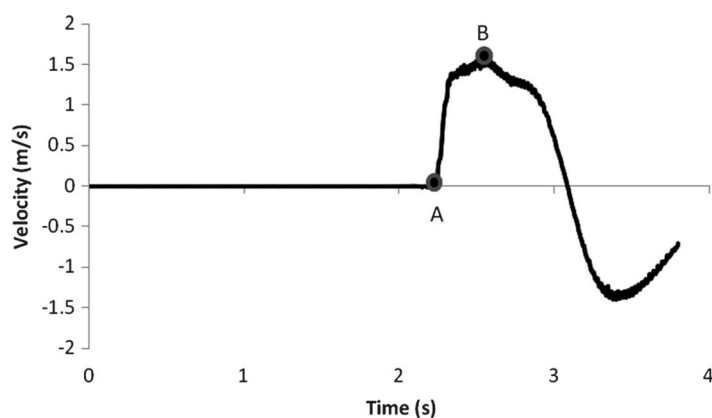


Figure 2. Typical velocity-time record obtained during the dynamic high pull at 30% of peak force. Point "A" represents the onset of rise in velocity while point "B" denotes maximum barbell velocity. High pull rate of velocity development was calculated as the slope of the segment between "A" and "B".

and time at 0 velocity (greatest knee flexion of countermovement), respectively.

RESULTS

The HPRVD correlated significantly ($p < 0.05$) with IsoRFD50 ($r = 0.52$) and IsoRFD100

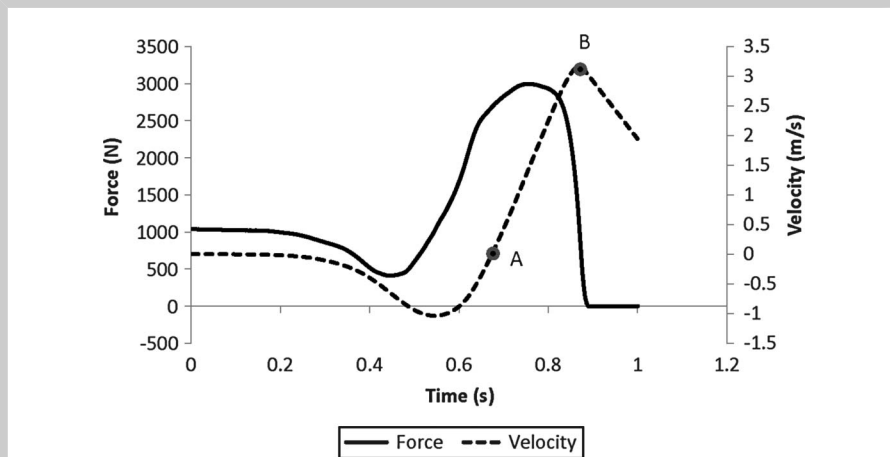


Figure 3. Typical force-time and velocity-time record obtained during a countermovement vertical jump. The slope of the segment between "A" and "B" represents vertical jump rate of velocity development. "A" indicates zero velocity and corresponds to the greatest knee flexion during the countermovement. "B" denotes vertical jump peak velocity.

TABLE 1. Pearson *r* correlations between force-time characteristics, velocity-time characteristics, and vertical jump performance.*

	Velocity-time and vertical jump performance				
	VJRVD	VJPV	VJHeight	HPPV	HPRVD
Isometric force-time					
IsoPF	0.14	0.38	0.38	0.03	0.01
IsoPF/BM	0.18	0.62†	0.61†	-0.60†	-0.31
IsoRFD50	-0.22	-0.35	-0.32	0.56†	0.52†
IsoRFD100	-0.16	-0.32	-0.29	0.56†	0.49†
IsoRFD150	0.39	0.39	0.41	0.16	0.22
IsoRFD200	0.35	0.32	0.31	0.17	0.27
IsoRFD250	0.43	0.40	0.40	0.25	0.25
IsoRFDMax	0.42	0.18	0.16	0.18	0.13
Dynamic force-time					
HPPF	-0.07	0.03	0.05	—	—
HPPF/BM	0.02	0.38	0.38	—	—
HPRFDMax	0.20	-0.25	-0.22	—	—

*IsoPF = isometric peak force; IsoPF/BM = isometric peak force relative to body mass; IsoRFD = isometric rate of force development at various time frames; HPPF = high pull peak force; HPPF/BM = high pull peak force relative to body mass; HPRFDMax = high-pull maximum rate of force development.

†Significant correlation ($p < 0.05$).

($r = 0.49$) (Table 1). The HPPV correlated significantly ($p < 0.05$) with IsoPF/BM ($r = -0.60$), IsoRFD50 ($r = 0.56$), and IsoRFD100 ($r = 0.56$) (Table 1). The VJHeight correlated significantly ($p < 0.05$) with IsoPF/BM ($r = 0.61$), whereas VJPV correlated significantly ($p < 0.05$) with IsoPF/BM ($r = 0.62$) (Table 1). High pull velocity-time characteristics did not

significantly correlate with any other isometric force-time variables (Table 1). In addition, vertical jump parameters (VJRVD, VJPV, VJHeight) did not significantly correlate with any other isometric or dynamic force-time variables (Table 1). The IsoRFDmax and HPRFDMax occurred in time windows of 0.79 ± 0.38 seconds and 0.23 ± 0.09 seconds, respectively. The VJRVD and HPRVD occurred in 0.27 ± 0.04 seconds and 0.30 ± 0.03 seconds, respectively.

For the significant correlations between high pull velocity-time characteristics and isometric force-time characteristics, we obtained the following 95% confidence intervals: HPRVD and IsoRFD50: 0.09 to 0.79; HPRVD and IsoRFD100: 0.05 to 0.77; HPPV and IsoPF/BM: 0.20 to 0.83; HPPV and IsoRFD50: 0.14 to 0.81; HPPV and IsoRFD100: 0.14 to 0.81. For the significant correlations between vertical jump parameters and isometric force-time characteristics, we obtained the following 95% confidence intervals: VJHeight and IsoPF/BM: 0.22 to 0.83; VJPV and IsoPF/BM: 0.23 to 0.84.

DISCUSSION

This study aimed to determine the relationship between dynamic exercise velocity-time parameters, force-time characteristics of isometric and dynamic muscle actions, and vertical jump performance. High pull kinematics including rate of velocity development and peak velocity shared similar correlational profiles, demonstrating relationships with explosive isometric force production 50 and 100 milliseconds from the onset of contraction ($r = 0.49$ – 0.56). These results point to a present, albeit limited, role of explosive strength as indicated by the rate of rise in force 50 and 100 milliseconds after contraction initiation on the ability to quickly attain high barbell

onstrating relationships with explosive isometric force production 50 and 100 milliseconds from the onset of contraction ($r = 0.49$ – 0.56). These results point to a present, albeit limited, role of explosive strength as indicated by the rate of rise in force 50 and 100 milliseconds after contraction initiation on the ability to quickly attain high barbell

velocities. In other words, force production shortly after the onset of contraction accounts for approximately 24–31% of the variance in HPRVD and HPPV.

Support for the potential influence of explosive strength 50 to 100 milliseconds from the onset of contraction on barbell acceleration may be deduced from the time that corresponded with the greatest HPRVD value. The HPRVD occurred in 0.30 ± 0.03 seconds (~ 300 ms), and, therefore, the neuromuscular system with a propensity for rapid force expression at preceding times would logically best influence barbell acceleration. This study found HPRVD to correlate significantly with IsoRFD50 and IsoRFD100, lending support to the aforementioned concept. Although isometric RFD obtained at other times still occurred before 300 milliseconds (i.e., IsoRFD150–250), its expression early in the movement possibly holds greater importance because inertial resistance must be overcome for any barbell displacement to occur. Therefore, resistance exercise designed to enhance explosive strength, particularly 50 to 100 milliseconds after the rise in force, may improve implement acceleration up to peak velocity. Identification of specific exercises that augment explosive strength at these times would be a worthwhile component of future investigations.

Interestingly, HPPV correlated negatively with IsoPF/BM ($r = -0.60$), indicating that individuals containing greater force per unit body mass may exhibit a diminished capacity to produce high barbell velocities with a 30% IsoPF load. This finding lacks strong support from previous research documenting correlates between relative strength and high pull barbell velocity. Haff et al. (10) did not observe body mass adjusted mid-thigh pull isometric peak force to relate to peak velocity attained during the same exercise performed dynamically at 30% IsoPF in national level female weightlifters. Furthermore, the subjects with the greatest and least relative isometric peak force attained the second highest and fastest barbell peak velocities, respectively. This simultaneously refutes and validates the suggestion that individuals with greater body mass adjusted strength as measured by the mid-thigh isometric pull produce lower barbell velocities. The observed negative correlation possibly resulted from the inability of recreationally trained participants in the current study to generate the requisite absolute force to overcome barbell inertia and generate high velocities (assuming strength in absolute and relative terms relate inversely with each other). It should be noted however, that because of our relatively small sample size, the confidence limits are moderately large for this correlation, thereby reducing generalizability.

Force-time characteristics of isometric muscle actions also related to whole-body performance. The VJHeight correlated significantly with IsoPF/BM ($r = 0.61$) but not with IsoPF, HPPF, or HPPF/BM, suggesting body mass adjusted strength rather than absolute force production influence whole-body displacement in the vertical direction. Although the nonsignificant correlation between VJHeight and

HPPF/BM may suggest otherwise, the greater complexity of the dynamic high pull exercise compared with an isometric pull may be a confounding factor because one might expect varying degrees of proficiency by recreationally trained subjects relative to explosively trained athletes. Indeed, a previous study observed a significant relationship between countermovement jump height and body mass adjusted power clean 1 repetition maximum (1RM) but not absolute 1RM in collegiate football players and track and field athletes (18). With respect to our results regarding relative strength and jump height, the existing literature both supports (18,22) and refutes (10,28) our results. These discrepancies may have manifested by way of methodologic differences. Young et al. (28) assessed relative strength using an isometric squat, and it could be argued that this test would display less transfer to jump performance than the mid-thigh isometric pull. In other words, segmental contributors (e.g., knee extensors, trunk extensors, plantar flexors) and kinematic characteristics (body position, sequence of joint actions) of the countermovement vertical jump with arm swing from the start of the concentric phase bears greater resemblance to those of the mid-thigh isometric pull compared with an isometric squat.

Support for this argument can be found in an investigation by Nuzzo et al. (18), who examined the relationship between 4 measures of relative strength and jump height. All relative strength variables including a mid-thigh isometric pull analogous to the one used in the current study correlated significantly with jump height except isometric squat peak force. Furthermore, the disagreement with Haff et al. (10) in spite of a comparable isometric strength test could be the outcome of different jump assessments. Subjects in our study used an arm swing as opposed to hands on the hips, and the enhancement of lower-extremity kinetics using an arm swing parallels the upper-extremity augmentation of lower-body exercises incorporating pulling motions such as the mid-thigh pull. The use of a weightlifting sample in Haff et al. (10) provides a degree of support to this notion because these athletes actively use upper-body musculature during the second pull of competition lifts and may not be accustomed to arm restrictions.

In addition to its relationship with VJHeight, body mass adjusted isometric force expression also correlated significantly with VJPV ($r = 0.62$), indicating that this measure of relative strength accounts for approximately 38% of the variability in the aforementioned jump parameter. In contrast with this finding, Nuzzo et al. (18) did not find countermovement jump peak velocity to significantly relate to IsoPF/BM. Few investigations have focused on vertical jump peak velocity after a resistance training intervention to clarify this disagreement. In one particular study, 3 months of explosive jump training resulted in significant pre- to postcountermovement jump peak velocity improvements (6). Under the premise that this training modality improves relative strength, the observed relationship between VJPV

and IsoPF/BM appears logical. Greater amounts of research describe the benefits of resistance training on sprint velocity (7,20,21). On the basis of the movement of body mass inherent to sprinting and jumping, although in different planes, one might propose similar effects on vertical velocity.

Unlike the significant correlations between high pull kinematics and IsoRFD50–100, the latter force-time characteristic did not relate to any vertical jump parameters. Several studies did not observe significant correlations between isometric and dynamic RFD and vertical jump displacement (10,12,18). Paasuke et al. (19), however, reported strong, significant correlations between isometric knee extensor RFD and jump performance ($r = 0.62$ – 0.83), whereas Viitasalo et al. (27) demonstrated similar results. A possible explanation for this discrepancy could be the time period associated with rate of force development calculation because different explosive movements occur within various time frames (1). Because the execution of a vertical jump occurs in roughly 250 milliseconds (3,13), the strongest correlations with rate of force development might be expected in similar time periods; however, this does not appear so with respect to our results.

A novel aspect of this study that merits further discussion involves dynamic exercise rate of velocity development assessment. This measure of acceleration from 0 to peak velocity in the vertical jump did not correlate with any force-time characteristics. These results suggest a limited or nonexistent role of force parameters on whole-body acceleration in the vertical direction; however, this should be interpreted cautiously because previous research does not substantiate these suppositions. Cormie et al. (6) analyzed countermovement jump velocity-time curves after 12 weeks of jump squat training and observed increased, although not significant, acceleration.

Additional training studies exploring the effect of resistance exercise on whole-body acceleration in activities other than jumping support the positive effect of explosive conditioning (14,23). Harrison and Bourke (14) examined the effect of resisted speed training using a loaded sled on sprint acceleration. Postintervention assessment revealed greater acceleration from a static start to 5 m during a 30-m sprint. Spinks et al. (23) also examined the training effect of weighted sled towing on 15-m sprint acceleration and found an 8% pre- to postincrease, particularly in the 0- to 5-m interval. This form of resistive, explosive conditioning designed to augment rapid force production proved beneficial to sprint acceleration. This, in turn, refutes the implications expressed by our correlations between explosive force production and whole-body acceleration. Therefore, the lack of significant relationships between dynamic and isometric force-time measures and VJRVD should not be interpreted as a limited role of force on body mass acceleration.

PRACTICAL APPLICATIONS

This study demonstrated significant relationships between high pull velocity-time parameters and explosive force production 50 and 100 milliseconds from the onset of

contraction. Although exercises beneficial to rate of force development at these specific time frames require identification, our results suggest that training to develop explosive strength in the early phase after contraction initiation may enhance implement acceleration and velocity. Specifically, the greatest transfer of explosive training effects might be expected in activities such as weightlifting, which closely resemble the high pull assessed in the current study. Furthermore, maximizing relative strength may positively influence the capacity for body mass displacement in the vertical direction as required by athletes participating in jumping events.

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